

Unit 3: Nanotechnology and Instrumentation

Nano Materials

①

→ Nano = Extremely small (dwarf)
Nano → Greek = "dwarf" (unit of measurement)

→ Nano materials:-

Definition:- Nano materials are chemical substances or materials with at least one of its ^{external} dimensions are in the nano scale ^(1-100 nm) (or) range and exhibit properties different from its bulk form.

(or)
Materials reduced to the nano scale (below 100 nm) can show different properties compared to that they exhibit on macro scale, enabling unique applications are known as 'nanomaterials'.

→ Nano materials refers to the material whose scale length, in any dimension, is approximately

1 to 100 nm.
→ The size of a nano material varies from '1 nm to 100 nm'.

The nanometre (nm) is a unit of length in the metric system, equal to 'one billionth of a meter' (10^{-9} m (or) 0.000 000 001 m).

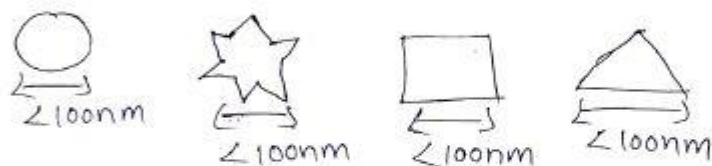
→ Every nanomaterial has three-dimensional structure of nano scale.

→ Nano refers to nanotechnology, or phenomena occurring on a scale of nanometres (nanoscopic scale).

→ Nano materials can be 'metals, polymeric materials or composite materials, ceramics, etc.

A particle of any shape with a dimension ranging from '1nm to 100nm'.

Ex:

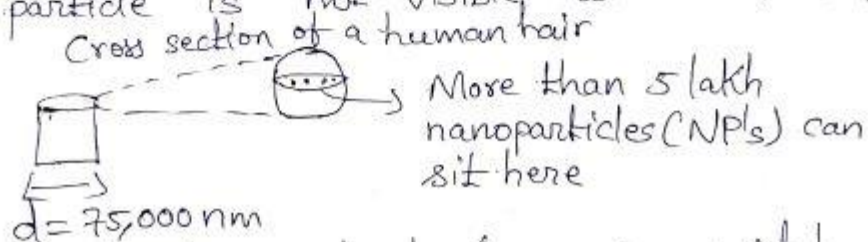


→ To understand how small 1 nm is, a few comparisons:

1) Water molecule is almost 0.3 nm across.

2) A red blood cell (RBC) is approximately 7000 nm wide.

3) Nano particle is not visible with naked eye.



'Human hair' is about '80,000 nm wide'.

*** Synthesis of Nanomaterials :-

There are two general approaches to the synthesis of nanomaterials and the fabrication of nanostructures.

1) Top-down approaches 2) Bottom-up approaches

1) Top-down approaches:-

In top-down approaches, bigger materials are broken down into small nano-size materials by means of various physical methods.

The main physical methods used for this purpose are pulsed laser deposition, nanolithography, etc.

2) Bottom-up approaches:-

In bottom-up approaches, desired nanomaterial is built by assembling of growth species like atoms, or ions, or molecules.

Ex:- Synthesis of polymers by connecting individual monomers, the growth of single crystals, deposition of films in electronic industry and production of industrial chemicals.

Few widely used bottom-up approaches are:-

There are two methods for the synthesis of nanomaterials:

- 1) Solution combustion method
- 2) Hydrothermal method

1) Solution combustion method

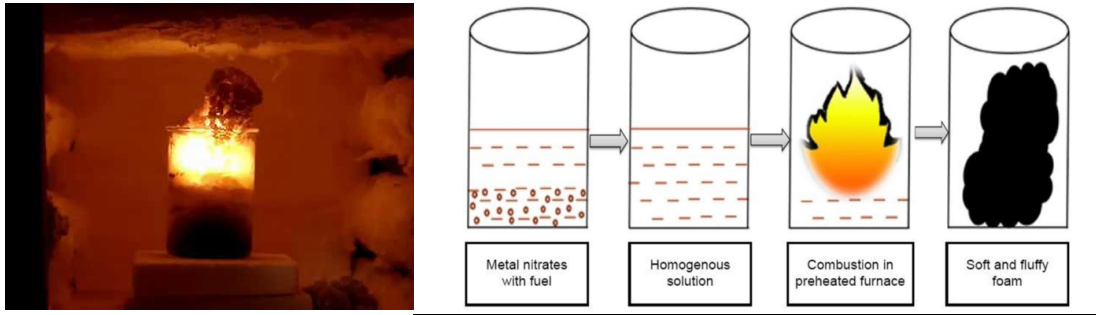
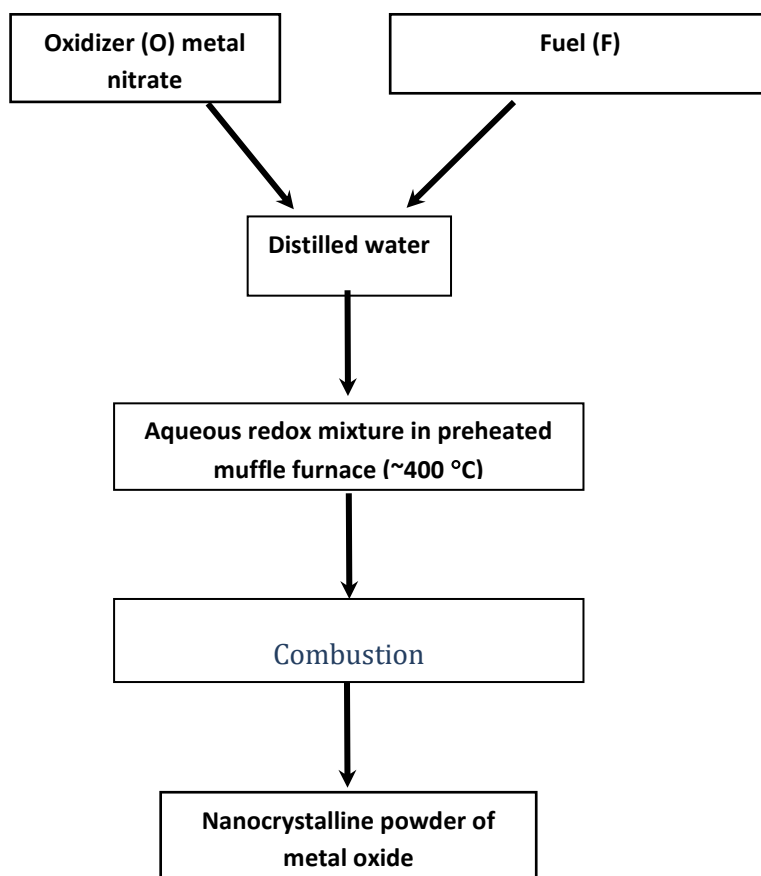


Figure: Solution combustion process and the synthesized nanomaterial formed inside the muffle furnace **(not required for exam and just for understanding)**

Q. What is solution combustion synthesis? Mention the advantages of this method. Explain synthesis of nano ZnO particles using solution combustion method.

Solution combustion process is a self-propagating high temperature synthesis in which exothermic reaction takes place accompanied by the evolution of gases.



Flow – Chart for synthesis of nano particles by solution combustion method

Procedure

It is another wet-chemical method like sol-gel and hydrothermal, which has been proved to be an excellent technique for preparing several grams micro/nanocrystalline materials due to its short processing time, low processing temperature. This technique is based on exothermic redox reactions that undergo self-sustaining combustion. Mixtures of oxidizer (usually metal nitrates) and a fuel (organic compound containing carbon and hydrogen; e.g. urea, citric acid or glycine as fuel) undergo spontaneous combustion under heating and the chemical energy from the exothermic reaction heats the precursor mixture to high temperatures. Such a high temperature leads to formation and crystallization of nano materials).

Advantages of solution combustion method:

Among wet chemical routes, the low temperature ***solution combustion process*** is an attractive one and finds several advantages like-

- The synthesis temperature (300 °C) employed is lower than those currently used in conventional route.
- Simple and energetically attractive process.

- Ability to dope desired amounts of impurity ions with better homogeneity.
- The powders are voluminous, foamy, sinteractive and in nanoscale having large surface area.
- Short reaction time (few seconds to 5 minutes)
- Chemical homogeneity

Example: Synthesis of ZnO nano particles by solution combustion method: **Refer lab manual**

2) Hydrothermal method for the synthesis of nanomaterial's



Figure: Different parts of Hydrothermal autoclave synthesis reactor **(not required for exam and just for understanding)**



Figure 2: Chemical reaction through Hydrothermal synthesis method **(not required for exam and just for understanding)**

Principle

In the hydrothermal synthesis, reactants are taken in water and heated at high temperatures (above 100 °C) and pressures (above 1 atm) in a closed vessel. A sealable Teflon-lined container is called a bomb (or) autoclave which is used to keep the reactants and water inside. Since it is operating at a very high temperature and high pressure, hence it is also called as the hydrothermal bomb.

Procedure

The crystal growth is performed in an apparatus from consisting of a steel pressure vessel called autoclave in which precursors is supplied along with water. After sealing, the autoclave is used to keep at high temperatures inside an oven. High temperature and pressure provides unique conditions for the solution to attain super saturation. Growth of nucleus is initiated from super saturated solution. Further, aging of the reaction mixture produces particles of well-defined morphology (a particular form, shape or structure). The product obtained is filtered, washed with water, air dried and finally, subjected to calcination (heating to high temperatures in the presence of air or oxygen).

The Hydrothermal autoclave synthesis reactor used to carry hydrothermal reaction at high pressure and high temperature.

Hydrothermal synthesis has been used to synthesize a variety of nanomaterials like nano-zeolite, nano metal oxides (Ex: TiO_2 , MnO_2 , etc).

Example:

Preparation of Bismuth ferrite (BiFeO_3) by hydrothermal method

Equimolar mixtures of bismuth nitrate ($\text{Bi}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$) and iron nitrate ($\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$) are dissolved in distilled water. Add 8M sodium hydroxide (NaOH) and stir for 30 min and poured into an autoclave. The hydrothermal treatment is conducted at 180°C for 12 h. The produced powder is collected at the bottom of the autoclave after cooling to room temperature. The

product is washed several times by repeated cycles of centrifugation in distilled water and ethanol. The obtained powder is heated at 80°C for 1 hour.

Applications of Hydrothermal Bomb (not required for exam and just for understanding)

- 1) Chemical Synthesis of Nano materials
- 2) Cultured crystal growth
- 3) Catalyst synthesis
- 4) Crystallization process
- 5) Hydrothermal oxidation
- 6) Petrochemicals
- 7) Sterilizing processes in the medical care industry
- 8) Microbiology
- 9) Medical Science
- 10) Dentistry
- 11) Polymerization reaction

Q - Mention the various techniques used in characterization of nano powder sample

The various characterization techniques used are – (i) Powder X-ray diffraction (ii) Scanning electron microscopy (SEM) (iii) Transmission electron microscopy (TEM)

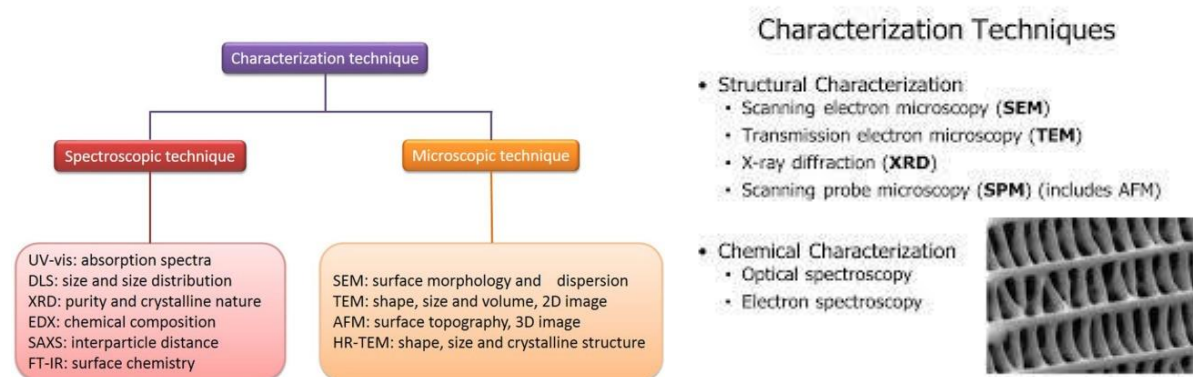


Figure: Characterization techniques for Nano materials

Applications of Nano materials

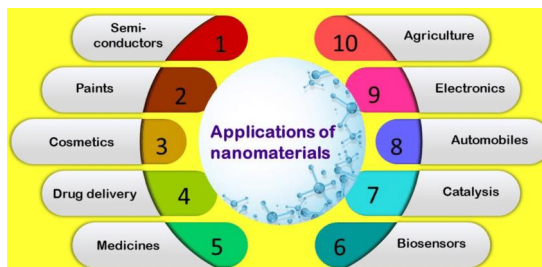
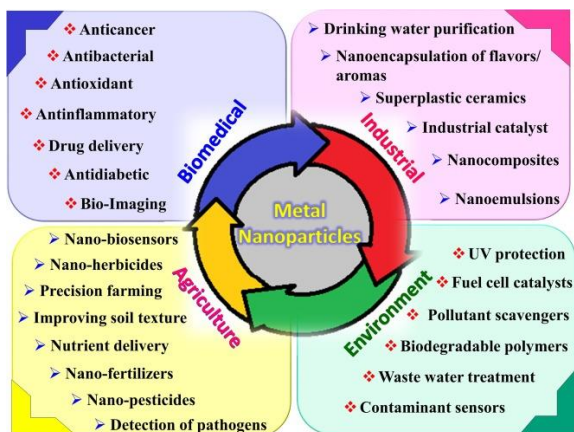
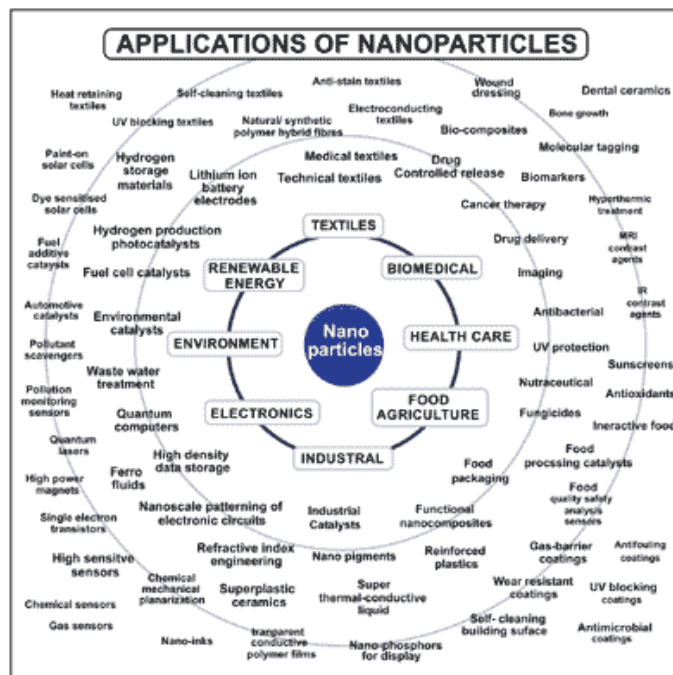


Figure: Applications of Nano materials

Application (field)	Examples
Medicine	Diagnostics, Drug delivery, Tissue engineering, Cryonics
Information and communication	Memory storage, Novel semiconductor devices, Electronic devices, Displays, Quantum computers

Heavy Industry	Aerospace, Catalysis, Catalysis, Construction Vehicle manufacturers
Consumer goods	Foods, Household, Optics, Textiles, Cosmetics, Sports
Environment	Water treatment: The removal of color, heavy metal, fluoride, etc Nano materials have better photo catalytic activity than their bulk counterparts due to very high surface area. Nano photo catalysts like ZnO, TiO₂ and CuO are used to degrade organic dyes and other impurities from waste industrial waste water.

Instrumental methods of analysis

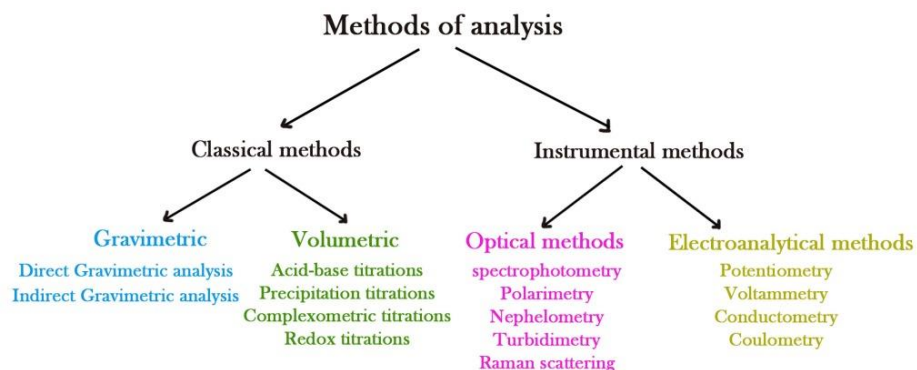


Figure: Representation of methods of analysis through instrumentation (not required for exam and just for understanding)

Introduction to analytical methods and instrumental method of analysis

- **Analytical chemistry** is a branch of chemistry that deals with chemical analysis. It also deals with the qualitative and quantitative characterization of a material or materials. Analytical chemistry is the study of the separation, identification, and quantification of the chemical components of natural and artificial materials. Analytical chemistry deals with the “qualitative and quantitative characterization of materials.
- **Qualitative analysis** is used to determine the identity of chemical species in the samples, whereas **quantitative analysis** is used to determine the amount or concentration of species present in samples. Two steps of chemical analysis in the characterization of matter or identification (qualitative analysis) which gives one or more components of the sample and estimation (quantitative) which gives exact quantity of constituents of a substance.
- **Instrumental methods** are mostly physical methods which use a simple or advanced instrument to measure physical quantities of the analyte by relating the concentration with light absorption, fluorescence, conductivity or potential. The physical methods have enabled the analyst to broaden the scope of analysis, since in many cases accurate measurements can be made without destruction of the sample. The instrumental methods have great advantage that it can investigate the structure of complex organic molecules, reaction kinetics, and even the biochemistry of living cells.
- **An instrumental method of analysis** is a physical property of a substance that are measured to determine its chemical composition. An instrument used for chemical analysis converts the property of the substance under investigation into a form that can be readily measured. The measured quantity is related to the quantity or quality of sample.

Example: Colorimeter measures the amount of radiation absorbed by a colored solution by converting the light radiation into electrical signal. Further the amount of light radiation absorbed is a function of concentration of the solution and the wavelength at which the absorption takes place, in the characteristic of the substance present.

Types of Instrumental methods of analysis

- Electrical methods (Electroanalytical)
- Optical methods (Spectroanalytical)

1) Electrical methods (Electroanalytical)

Electrical method of analysis involves the measurement of current, voltage or resistance in relation to the concentration of a certain species in solution. For example potentiometry, involves the measurement of the potential of an electrode in equilibrium with an ion to be determined.

Example: Potentiometry, conductometry

- #### **2) Optical methods** are based on, how the sample acts towards the electromagnetic radiation.

Colorimetry

Simple technique used to determine the concentration of compounds in a solution. This method is used for solution which are themselves coloured or which gives color with a suitable reagent. The intensity color of the solution is directly proportional to the concentration of the solution.

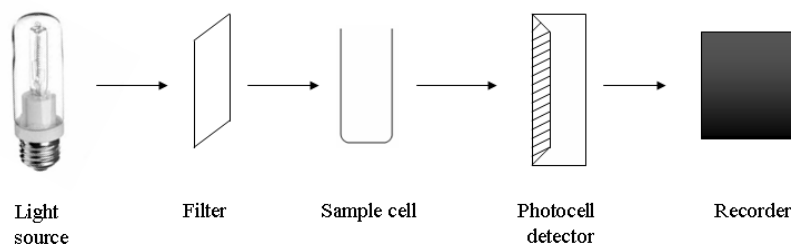
Theory

This method is based on Beer-Lambert's law. According to this law, when a monochromatic light is passed through a solution, part of light is absorbed by the solution. The extent of absorption depends on the concentration of the solution and on the path length of the light through the solution.

$$A = \epsilon C t$$

A= absorbance, C= concentration of the solution, t = path length, ϵ = molar absorption which is a constant at given wavelength.

Instrumentation



Photoelectric colorimeter consists of following parts

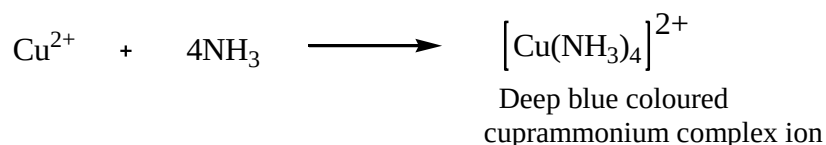
1. Tungsten lamps as light source
2. A filter which provides the desired wavelength range
3. A sample cell
4. A photocell detector

Light from tungsten lamp, after passing through the filter is allowed to fall on the solution taken in the sample cell. First, a blank solution is taken in the sample cell and placed in the path of light beam. Its absorbance is measured and it is adjusted to zero. Then the analyte solution is placed in the path of light and its absorbance is measured.

Example: Colorimetry is used in the estimation of copper in sample solutions

Principle

Solution containing Cu^{2+} ions is treated with NH_3 solution. Cu^{+2} ions react with NH_3 to form a deep blue coloured cuprammonium complex ion.

**Procedure**

Prepare 100 ml of 0.1 M solution of CuSO_4 . Transfer 2, 4, 6, 8 and 10 cm^3 of prepared CuSO_4 solution into different volumetric flasks from the burette. Add 2.5 cm^3 of ammonia solution to all the flasks and make upto to mark using distilled water. Stopper the flasks and shake well to get uniform concentration.

Prepare a blank solution by taking 2.5 cm^3 of NH_3 in volumetric flask, make upto the mark by using distilled water and shake well. Measure absorbance of all the solutions against blank at 620 nm using a photoelectric colorimeter.

To the given unknown solution, also add 2.5 cm^3 of NH_3 and make up to the mark with distilled water.

Then a calibration curve is obtained by plotting absorbance against concentration. From the calibration curve, the concentration of the Cu^{+2} in the solution can be determined.

Applications of colorimetry

1. Colorimetry is the most common analytical technique used in chemical and biochemical estimations in various laboratories including clinical laboratories.
2. It's the most widely used in clinical laboratories for the analysis of bio-molecules such as glucose, urea, creatinine, etc.
3. It is used in water and food testing laboratories to check the concentration of critical chemical parameters periodically.

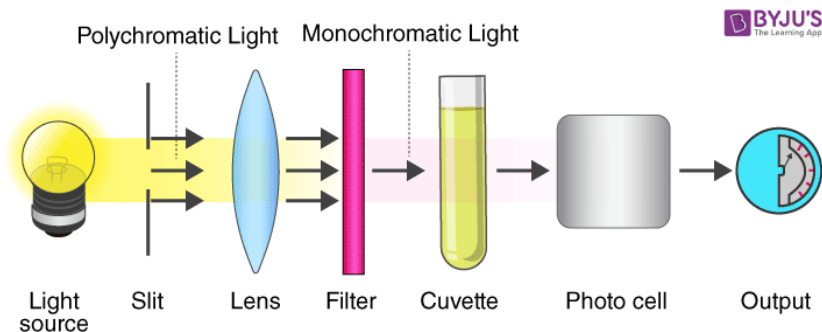


Figure: Colorimetry **(just for understanding and not required for the exam)**

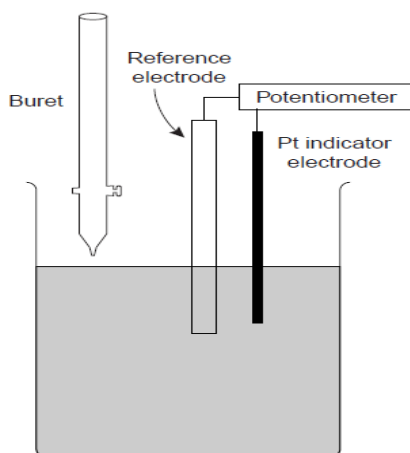
Potentiometry

The procedure of using measurement of e.m.f to determine the concentration of ionic species in solution is referred to as potentiometry. The relation between electrode potential and metal ion concentration is given by the Nernst equation:

$$E = E^{\circ} + \frac{0.0591}{n} \log [M^{n+}]$$

It can be seen from the equation that the potential of an electrode E depends upon the concentration of the ion M^{n+} to which it is reversible.

Instrumentation



Potentiometer consists of a reference electrode, an indicator electrode and a device for measuring the potential. The indicator electrode responds rapidly to the changes in the potential due to the concentration changes of the analyte.

A known volume of the analyte is taken in a beaker and its potential is determined by connecting the assembly to a potentiometer. The titrant is added in increments of 1 ml and the potential is measured each time. Close to the equivalence point, there is a sharp increase in the potential. The end point is determined by plotting change in potential against the volume of titrant.

Applications of potentiometry

1. It finds applications in oxidation-reduction titrations to estimate the concentration of analyte in the sample solution.
2. In redox titrations, the indicator electrode is an inert metal such as platinum and reference electrode (Saturated calomel electrode). The potential of the electrode depends on the concentration of the substance being oxidized or reduced.

$$E = E^{\circ} + \frac{0.0591}{n} \log_{10} \frac{[\text{Oxidized}]}{[\text{Reduced}]}$$

pH-sensor (Glass electrode)

Ion-selective electrodes:-

Definition:- The ability to respond only to certain specific ions in a mixture of ions and ignoring the other ions totally (or) completely.

In other words, the potential developed by an ion-selective electrode depends only on the concentration of that ions of interest.

Ex:- Glass membrane electrode is only H^+ ions selective. (16)

→ Ion-selective electrodes mainly consist of ion-selective membrane.

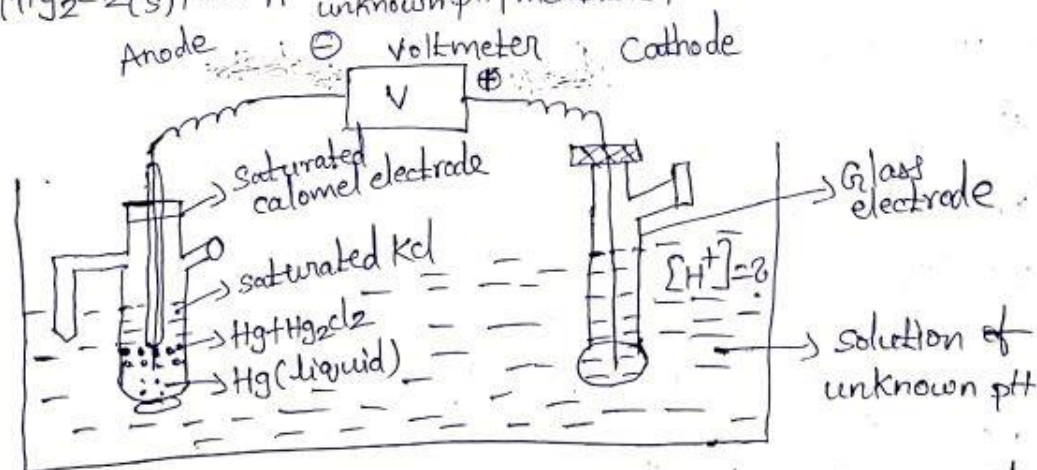
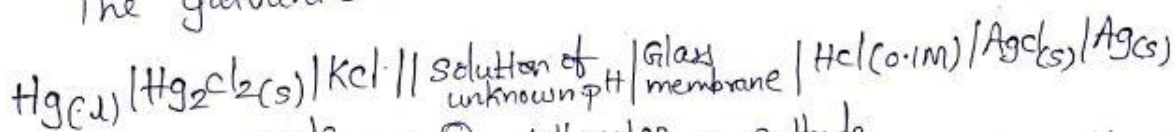
→ Ion-selective electrodes are used to measure the concentration of a particular cations or anions (H^+ , Na^+ , Li^+ , K^+ , Mg^{2+} , CN^- , NO_3^- , F^- , etc) in the mixture.

→ These electrodes are highly sensitive, selective and suitable for rapid, low cost, accurate analysis. Hence, they are currently used in many fields including clinical diagnostics, industrial process control, environmental monitoring and physiology.

Determination of pH of a solution using Glass electrode:

In order to determine the pH of any given solution, a galvanic cell is constructed by combining glass electrode (internal reference electrode) with standard calomel electrode (SCE, a reference electrode).

The galvanic cell obtained is represented as



The EMF of the complete cell when immersed in a solution of unknown pH i.e., pH is unknown (u).

$$E_{\text{cell}}(\text{Cu}) = E_{\text{right}} - E_{\text{left}} \quad (\text{Anode})$$

(or)

$$E_{\text{cell}}(\text{Cu}) = E_{\text{Glass}} - E_{\text{SCE}}$$

$$E_{\text{cell}}(\text{Cu}) = [L' - 0.0591 \text{pH}_{\text{Cu}}] - E_{\text{SCE}}$$

Rearranging this equation for pH.

$$\therefore \text{pH}_{\text{Cu}} = \frac{L' - E_{\text{SCE}} - E_{\text{cell}}}{0.0591} \quad \text{at } 298 \text{ K } (25^\circ \text{C})$$

Applications of pH

pH measurement is used in a wide variety of applications:

- 1) **Food and beverages: It is important to monitor pH levels in the production of any food or beverage product.**
- 2) **The pH level of soft drinks ranges from 2.1 to 7.4 depending on what kind of ingredients and additives are added.**
- 3) Agriculture
- 4) Wastewater treatment
- 5) Industrial processes
- 6) Environmental monitoring
- 7) Research and development
- 8) pH is a measure of the acidity or alkalinity of a solution
- 9) Aquariums and fish tanks

Prepared and compiled by

Dr. Sampath Chinnam, M.Sc., Ph.D., Post-Doc (South Africa)

Assistant Professor

Department of Chemistry

M.S. Ramaiah Institute of Technology

MSR Nagar

Bengaluru, Karnataka 560054

India